

Project Report on
**Structural Optimization of Composite Windmill Blade
with variation of Density**

Submitted in the partial fulfillment of requirements for the degree of
Bachelor of Engineering (Mechanical)

By

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CERTIFICATE

This is to certify that the project entitled “Structural optimization of composite windmill blade with variation of Density” is a bonafide work of “Shubham Shashank Palande” (Roll No. 4767) submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of “Undergraduate” in “B.E. Mechanical”.

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PROJECT REPORT APPROVAL FOR B.E

This project report entitled *(Structural optimization of composite windmill blade with variation of Density)* by (Pramanshu Rajesh Tiwari, Utkarsh Santosh Shilam, Shubham Shashank Palande, Rahul Kisan Raghatwan) is approved for the degree of *(B.E – Mechanical Engineering)*.

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DECLARATION

I declare that this written submission represents my ideas in my own words and where others ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all the principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

The project focuses on Design, Fabrication and Testing of a VAWT (vertical Axis Wind Turbine). The project is an ongoing research project where we tried to achieve a portable and fiscally responsible prototype. A major concern was fashioning the design to enable the VAWT to operate with maximum efficiency. Several parameters were analyzed with respect to wind speed to determine the best value for each parameter which would give the highest efficiency, thus ensuring the maximum ultimate performance of the VAWT. The parameters that were considered for analyzing are the number of blades the rotor should have, positioning of the blade (i.e. the distance from the shaft to blade and the angle the blade creates with the shaft), the shape of the blade, the angle of the blade and it' s density, so as to generate the highest efficiency. Above parameters were analyzed using ANSYS/Fluent software package and the ultimate design was produced in accordance with the obtained results. Results of every analysis are appended in this report

Keywords: - VAWT, Windmill, Blade density, Efficiency, etc.

ORGANIZATION OF DISSERTATION REPORT

This research work is organized as follows: -

Chapter 1 includes basic introduction of Vertical Axis Wind Turbine (VAWT). This chapter includes and discusses about VAWT and its harmful effects to surrounding, this chapter also includes problem statement, Motivation for work, Objectives and Limitations.

Chapter 2 highlights previous work on VAWT, the way of use, it includes the study of past methods which are used to harness energy from VAWT.

Chapter 3 describes the overall research methodology used to fabricate the device. Discussed how the capacity of device was decided.

Chapter 4 explains the design in detail. Design, Hardware and Software components are briefly described. Also, the working of device, the research and development cost and the final cost are mentioned and component level cost is also described

Chapter 5 lists the application and limitation of this research work.

Chapter 6 lists the future scope and concludes the presented research work.

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CHAPTER 1

INTRODUCTION

CHAPTER 1

INTRODUCTION

A vertical-axis windmill is a type of windmill where the main rotor shaft is set transverse to the wind while the main components are located at the base. A vertical axis windmill has its axis perpendicular to the wind streamlines and vertical to the ground. A more general term that includes this option is "transverse axis windmill" or "cross-flow windmill. An aerodynamically efficient blade is the prime necessity to extract maximum power from a windmill. With the increasing size of the windmill blade, the blades are now basically made of composite materials. Composite materials satisfy complex design constraints such as lower weight and proper stiffness, whereas providing good resistance to the static and aerodynamic loading. There are two major issues that have not yet been resolved satisfactorily in the optimization of the composite wind turbine blade. One of the issues is the aerodynamic load calculations, and the other one is the establishment of an optimal model of the composite wind turbine blade. There are two methods for the calculation of the aerodynamic load. One method is translating it into concentrated force based on classical momentum blade element theory. But this method cannot truly reflect the pressure distribution of the blade surface. Another method is the use of computational fluid dynamics; the pressure distribution is obtained with CFD software, and then loaded into the finite element model to investigate the mechanical strength of the composite wind turbine blade.

1.1 HISTORY

European style grain grinding windmills are what usually comes to mind when thinking of early-age wind power. However, even if these types of horizontal axis windmills were introduced in Europe no later than the 12th century, the first recordings of wind turbines are from the 9th century, describing Persian vertical axis windmills. Vertical axis windmills might have been in use in the Afghan highlands as early as the 7th century BC. These early VAWTs were simple devices based on aerodynamic drag rather than on aerodynamic lift. One side of the turbine was covered and the wind simply pushed the blades on the power-generating side of the turbine and thus created torque. Using the aerodynamic lift created by pressure difference due to the shape of the blade is far more efficient than using drag. The first lift-based vertical axis wind turbine was invented by Georges Darrieus in France during the 1920s (French patent in 1925; US patent in 1931). The Darrieus patent covers the "egg beater" shaped turbine with troposke in curved blades, sometimes referred to as a full-Darrieus or Φ -configuration Darrieus turbine but

is commonly just called a Darrieus turbine. The Darrieus patent also covers the straight-bladed H-rotor (or giro mill). The curved blades of the Darrieus turbine are mounted directly to the vertical torque tube (the rotating shaft, which also serves as the support structure) at the top and bottom. The top of the turbine can be held upright by guy cables or the torque tube can be cantilevered from the base. In contrast, the straight blades of the H-rotor are attached by a cross arm. The H-rotor can drive a generator placed either at the top of the tower or, using a shaft placed inside the tower, at ground level.



Fig: 1.1 VAWT

1.2 MOTIVATION

World is running out of fuels due to excessive amount of extraction of fossil fuels. Wind Turbine is a thing which can be placed in any place where wind is in good flow. Helpful for remote areas like hilly region like Himalayas, western ghats, central India (Dandakaranyan, Chota Nagpur Plateau,) NE region of India. It can be used in Industrial Region like MIDC or any other state industrial region

1.3 PROBLEM STATEMENT

One of the main drawbacks of this turbine was the air speed should be more than 2 m/s to rotate the wind turbine with deflectors. Because of that, in low velocities turbine rotation disturb by the deflectors due to inertia. So, one of the main modifications can be done to get rid of that problem is increasing the length of the deflector support. Then it won't disturb to the rotation of the wind turbine.

1.4 ESSENTIAL COMPONENTS OF VAWT

1.4.1 BLADE

The NACA airfoils are airfoil shapes for aircraft wings developed by the National Advisory Committee for Aeronautics (NACA). The shape of the NACA airfoils is described using a series of digits following the word "NACA". The parameters in the numerical code can be entered into equations to precisely generate the cross-section of the airfoil and calculate its properties.

NACA 2410 - NACA 2410 airfoil

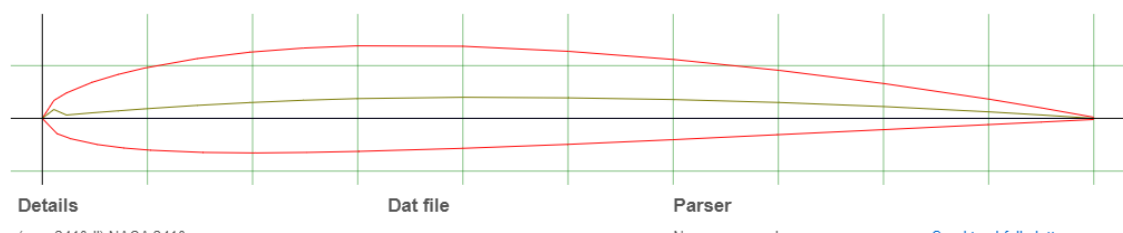


Fig: 1.2 Airfoil

(naca 2410-il) NACA 2410

NACA 2410 airfoil

Max thickness 10% at 29.9% chord.

Max camber 2% at 40% chord

1.4.2 HOUSING

To provide a smooth functioning of the wind blades we provided with a cylindrical casing with both of its end welded with bearings. It reduces the height of its center of mass also distributes its overall weight properly and makes a stable performance.



Fig: 1.3 Bearing Casing

1.4.3 SUPPORT STRUCTURE

To provide it stability and a height for better performance



Fig: 1.4 Steel Stool

1.5 CHALLENGES FACED

1.5.1 LESS ROTATIONAL EFFICIENCY

Vertical axis wind turbines often have less rotation efficiency. This is part of the reason why vertical axis wind turbines have lower efficiency. Due to the rotor design, not all the blades on the vertical axis rotor receive incoming wind at the same time. In fact, only the wind-facing blades are driven by the wind to turn while the others are simply following along. During rotation, vertical axis rotors are also faced with more drag— or aerodynamic resistance— on the blades. This is especially shown on Savonius wind turbines because they have wider blade surfaces.

1.5.2 LOWER AVAILABLE WIND SPEED

Since vertical axis wind turbines are typically installed on ground level, they do not harness higher wind speeds often found at higher levels. Consequently, less wind energy is available for ground-level vertical axis wind turbines. A common solution to this is to install the turbine on the rooftop of a building. To address this problem, we have improved the rotor design so that it can be mounted on top of a mast. The rotor and the mast combined tower our vertical axis wind turbine to 10 meter high, with the generator and power electronics located at the 4-meter height.

1.5.3 COMPONENT WEAR-DOWN

Often placed on ground level and populated environments, vertical axis wind turbines face more turbulence and issues of vibrations. When in operation, not only do the blades need to withstand more force, the bearing between the rotor and the mast also needs to endure higher pressure. In earlier models, it was more likely for the blades to bend or crack. For others, this can result in more maintenance and therefore more cost. To strengthen our own turbines, we have taken wear-down calculations into consideration for design and manufacture. The Luvside vertical axis wind turbines are now robust enough to stand wind speed equivalent to a strong tropical storm.

1.5.4 LESS EFFICIENCY

Vertical axis wind turbines are known to have less efficiency compared to horizontal axis wind turbines. This is mainly due to the nature of their design and operational characteristics. On

average, the efficiency of a horizontal axis wind turbine lays between 40 to 50 %, meaning the turbine is able to convert 40% to 50 % of the kinetic energy it receives into actual electrical power. On the other hand, a Savonius vertical axis wind turbine has an average efficiency of 10 to 17 %, while the Darrieus vertical axis wind turbine reaches 30 to 40 %. Even so, given the suitable environment, a Savonius wind turbine can still produce enough power to support annual consumption of a normal two-person household.

1.5.5 SELF-STARTING MECHANISM

While horizontal axis wind turbines and Savonius wind turbines start turning automatically once receiving wind, Darrieus wind turbines often rely on a starting mechanism. This is because the wing design of Darrieus wind turbines does not always guide the wind to form enough torque for rotation. At the moment, this is still a challenge that many, including us, are trying to tackle.

1.6 CONCLUSION

As an engineer it is our duty to address the above mentioned problem and try to find an overcome problem and give best results for humankind and safe guarding mother earth.

CHAPTER 2

LITERATURE REVIEW

CHAPTER 2

LITERATURE REVIEW

The chapter presents the review of past work on Incinerators, the way of disposal of waste. The literature has revealed the basic way to dispose the waste. The people worldwide are facing many challenges due to improper disposal of waste, it is difficult to clean the mess that is spread in the environment. It includes the study of past methods which are used to dispose waste.

2.1 EXISTING LITERATURE

Niranjana. S.J

Investigated the power generation by vertical axis wind turbine. In this paper the power is generated by fixing the wind mill on the road high ways. When the vehicle is passed through the road at high speed the turbine of the wind mill rotates and generates the power sources. This analysis indicates that the vertical axis wind turbine can be able to attain the air from all the direction and produces the power of 1 kilowatt for a movement of 25 m/s. The efficiency of vertical axis wind turbine can be increases by modifying the size and shape of the blade.

Abmjit N Roy

Analyzed the design and fabrication of vertical axis economical wind mill. This paper indicates that vertical axis wind mill is one of the most important types of wind mill. In this main rotor shaft is connected to the wind turbine vertically with the generator and gear box which can be placed near the ground. Performance characteristics such as power output versus wind speed or versus angular velocity must be optimized in order to compete with other energy sources which make the process economically and eco-friendly. The experimental result shows that wind turbine is placed on the top of the building in an ideal position to produces electricity. The power generation becomes easy and it is used for various applications such as street light, domestic purpose, agriculture etc.

D.A. Nikam

Analyzed the literature review on design and development of vertical axis wind turbine blade. This paper explains that the wind mill such as vertical and horizontal wind mill is widely

used for energy production. The horizontal wind mill is highly used for large scale applications which require more space and huge investment. Whereas the vertical wind mill is suitable for domestic application at low cost. The generation of electricity is affected by the geometry and orientation of the blade in the wind turbine. To optimize this by setting the proper parameter for the blade design. The experimental result indicates that the blade plays critical role in the performance and energy production of the turbine. The optimized blade parameter and its specification can improve the generation of electricity.

Altab Hossain

Investigated the design and development of A 1/3 scale vertical axis wind turbine for electrical power generation. In this paper the electricity is produce from the wind mill by wind power and belt power transmission system. The blade and drag devices are designed in the ratio of 1:3 to the wind turbine. The experiment is conducted by different wind speed and the power produced by the windmill is calculated. The experimental result indicates that 567 W power produced at the speed of 20 m/s while 709 W power produced at the speed of 25 m/s. From this, the power production will increases when the velocity is high.

M. Abid

Analyzed the design, development and testing of a Savonius and Darrieus vertical axis wind turbine. This paper shows that vertical axis wind mill is more efficient when compare to horizontal axis wind mill. The Darrieus turbine consists of 3 blades which can start alone at low wind speed. When Savonius turbine is attached on the top of existing wind mill which provide the self-start at low wind speed. The result indicates that the Darrieus vertical axis wind turbine acts as a self-starter during the testing. The function required the starting mechanism which can be provided by the combination of NACA 0030 aerofoil and Savonius turbine. The high blade thickness of the NACA 0030 aerofoil will improves the self-starting capability of the turbine.

ParthRathod

Analyzed a review on combined vertical axis wind turbine. In this paper, the increased efficiency is achieved based on the characteristics such as aspect ratio, tip speed ratio, velocity and other geometry parameter. The experiment is conducted to increase the power production and efficiency of a wind turbine. The development of design is optimized by combining the blade structure and the flow performance. The result indicates that the efficiency of turbine is

always based on the wind speed and climatic conditions. The lowest aspect ratio improves the power coefficient of the turbine. The power generation of combined rotor is high compare to the single Savonius and Darrieus rotor.

KunduruAkhil Reddy

Investigated a brief research, study, design and analysis on wind turbine. This paper evaluates the aerodynamic performance of variable speed fixed pitch horizontal axis wind turbine blade using two and three dimensional computational fluid dynamics. The primary objective of the paper is to increases the aero dynamic efficiency of a wind turbine. The blades are designed using different type of airfoils which are associated with angle of attack. The blade design is responsible for the efficiency of the wind turbine. The design of the blade is done using Q- blade software. The result indicates that the power output is determined using blade elemental theory. The power output of designed blade design is higher when compare to existing design of the blade.

Piyush Gulve

Analyzed the design and construction of vertical axis wind turbine. This paper indicates that vertical axis wind turbine is more efficient than horizontal axis wind turbine because it requires compact space for producing same amount of electricity and less noise. The result of the paper indicates that the efficiency of wind turbine may reduce due to manufacturing error and frictional losses. From the above literature review, it is clearly understood that the efficiency of wind turbine is always based on the parameters such as design and size of the blade, aspect ratio, tip speed ratio, blade angles and velocity. The power production of combined vertical and horizontal wind mill is high compare to vertical axis wind turbine and horizontal axis wind turbine. It requires less space for high generation of electricity.

2.2 CONCLUSION

The combined vertical and horizontal axis wind turbine increases the efficiency and production volume while compared with separate vertical and horizontal axis wind turbine. This will reduce the area required for the installation of wind turbine by fixing the vertical and horizontal wind turbines in a single tower. It will accumulate more number of wind towers at less area compared to VAWT and HAWT. The implementation of combined vertical and horizontal axis wind turbine will solve the issue on the usage of fossil fuels and highly helpful for the environment to safeguard from global warming.

CHAPTER 3

METHODOLOGY

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METHODOLOGY

3.1 INTRODUCTION

Requirements. For the objectives of the current study to be feasible, the requirement for a straightforward, modular, and automated design framework became realized. Successfully taking a physical system, such as a wind turbine, and attempting to adjust, analyze, and optimize the design to satisfy an objective, or multiple objectives, required more than a few bundled files and programs requiring user input. In fact, it was this realization that sparked the idea of a simple, automated optimization methodology. The optimization methodology proposed is a unique and modular system aimed at relaxing the ties between the computer and operator and simplifying the design process so more time can be spent analyzing a solution and understanding the physics of the problem.

Unique Modular Design. A modular system is one in which entire parts of the system can be removed and replaced without compromising the process flows within the system. Therefore, for a module to be removed and replaced, it must be substituted with a module of equivalent functionality. The first step in the optimization process is generating the geometry. This geometry is described as a set of Cartesian coordinates and is passed to the mesh generation module. This tool is used to discretize the fluid domain and output a specific file to the CFD solver module. This module calculates a solution and passes information to the post processing module. The post processing tools manipulate the data, calculate the objective function value, and pass it to the optimizer. If the objective function is considered a maximum value, the optimization terminates. If not, the process starts over. Details pertaining to each of the modules used in this work will be discussed in the next section.

3.2 METHODOLOGY

Another tough task for us was selecting the material of the casing. So, we decided to go with mild steel and ceramic coating on it. After conducting some tests, we also decided to eliminate the intake fan and use bowers. It also helps us in reducing the electricity consumption and make the device compact.

CHAPTER 4

DESIGN OF BLADE AND VAWT

CHAPTER 4

DESIGN OF BLADE AND VAWT

4.1 PREVIOUS DESIGN

There were five designs before confirming the final design for manufacturing. The prior designs were bulky and not efficient placement of components was also not convenient. To make the design compact, it was decided to use only Aluminum sheet. Due to high durable, light in weight, low costing, less rigidity.

Table 4.1 Materials

Materials used	Density
Galvanized Iron	7.809g/cm ³
Sun board	0.33g/cm ³
Aluminum Composite	1.51g/cm ³
Aluminum	2.7g/cm ³

To achieve results of motion we use motor and checked air speed in Kharghar .Which gives optimal results.

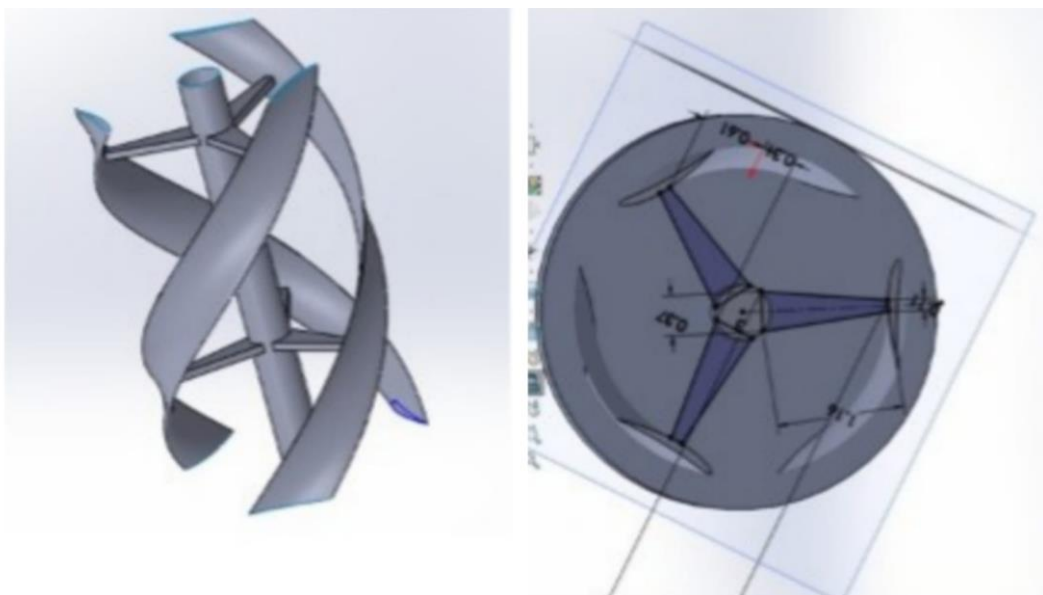


Fig: 4.1 Previous CAD design of VAWT

4.2 FINAL DESIGN

After various optimization final optimized proposed model was adopted. The design offers better portability, compactness, simple design and better efficiency.

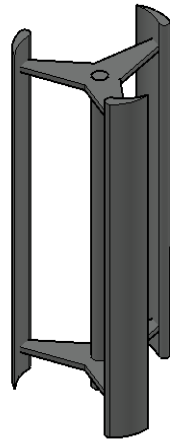


Fig: 4.2 Final design of VAWT

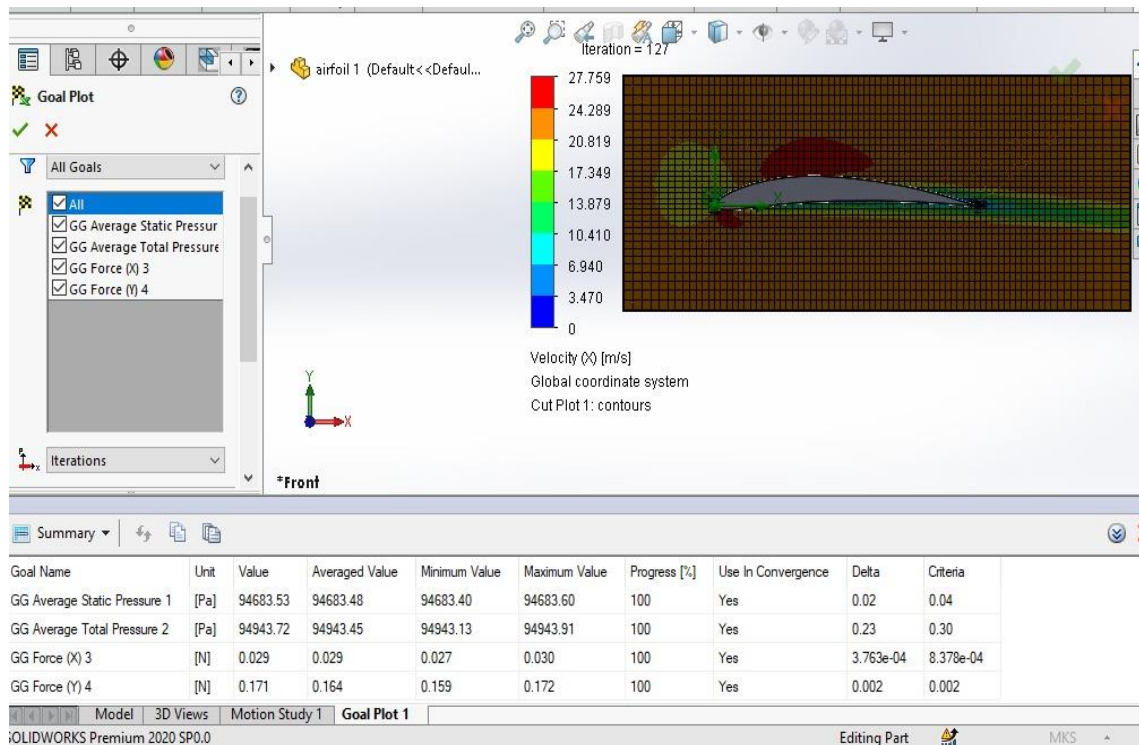


Fig: 4.3 Solidworks analysis with Solidwork Simulation

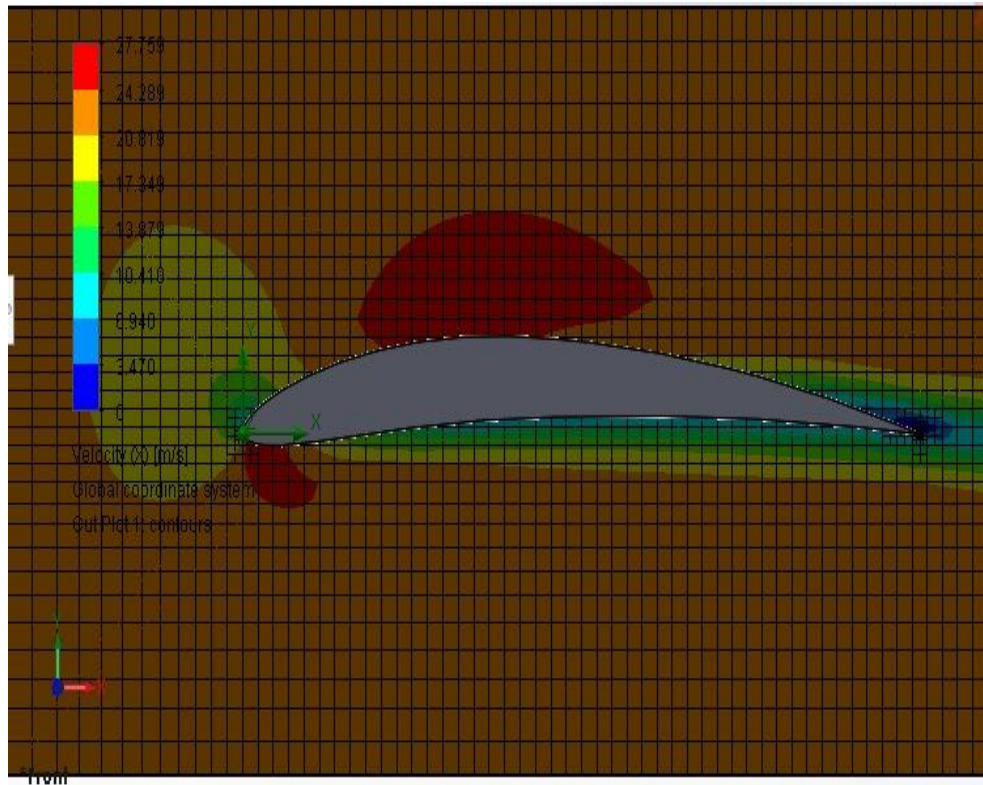


Fig: 4.4 Solidwork Simulation

4.3 NUMERICAL

Power Output Formula: $P = \frac{1}{2} \rho A_w v^3 \eta C_P$

Where ρ = Air Density

A_w = Area Swept

v = Wind Speed

η = Transmission Efficiency

C_P = Power Coefficient

- Amongst the above variables only air density, wind speed and area swept are arbitrary other two variables are obeying boundary conditions

- Air Density changes with changes in temperature and with the varying pressure related changes for plain sailing we assume average air density at average temperature of 30°C 1.165 Kg/m³.
- Average Wind speed in Navi Mumbai are is 3.7m/s with maximum wind speed being 11m/s.
- Transmission Efficiency is assumed to be at 59.9% and the power coefficient is 0.593 maximum in HAWT this is also known as Beltz Limit, for VAWT the maximum CP is 0.4. We will assume that to 0.15 after considering various values from the research we concluded.
- $P = \frac{1}{2} \rho A w v^3 \eta CP$

Power output per unit swept area

$$P = (0.5 * 1.165 \text{ Kg/m}^3 * 1 \text{ m}^2 * (3 \text{ m/s})^3 * 59.9 * 0.15) 1000$$

$$= 0.1413 \text{ Kw/h}$$

Since there are 8760 hours per year so the 0.1413 kw/h becomes 1237 Kw/h per year

- Below table shows the power output at different velocities produced by 59.9% efficiency and 1m² wind turbine in an ideal condition

4.4 ACTUAL MODEL



Fig: 4.5 Support Structure



Fig: 4.6 Bearing Casing



Fig: 4.7 Blades

4.5 COST ESTIMATION

Cost estimation may be defined as the process of forecasting the expenses that must incurred to manufacture a product. These expenses take into a consideration all expenditure involved on design and manufacturing with all related services facilities such as pattern making, tool making as well as portion of the general administrative and selling costs.

4.5.1 PURPOSE OF COST ESTIMATING

- To determine the selling price of the product for a quotation or contract so as to ensure a reasonable profit to the company.
- Check the quotation supplied by vendors.
- Determine the most economical process and material to manufacture the product.
- To determine standards of production performance that may be used to control the cost.

Basically, the budget estimation is of two types:

1. Material cost.
2. Machining cost.

4.5.2 MATERIAL COST ESTIMATION

Material cost estimation gives the total amount required to collect the raw material which has to be processes or fabricated to desire size and functioning of the components. These materials are divided into two categories:

1. Material for fabrication:

In this the material is obtained in raw condition and is manufactured or processes to finished size for proper functioning of the component.

2. Standard purchased parts:

This includes the parts which was readily available in the market like screws etc. A list forecast by the estimation stating the quality, size and standard parts, the weight of the raw material and cost per kg for the fabricated parts.

4.5.3 MACHINING COST ESTIMATION:

The cost estimation is an attempt to forecast the total expenses that may include to manufacture apart from material cost. Cost estimation of manufactured parts can be considered as judgment on and after careful consideration which include labor, material and factory services required to produce the required part.

Cost estimation is done as follows:

Cost of project = **(A)** Material Cost + **(B)** Machining Cost + **(C)** Testing Cost

(A) Material Cost is calculated using:

1. Raw material cost.
2. Finished process cost

1. Raw Material Cost:

It includes the material in the form of the material supplied as round bars, angles, square tools, plates along with the strip material form, we have to search for the suitable available material as per the requirements of designed safe values.

This device was built upon trial and error method, trial and error method required of buying new components conduct experiments on the bought components and observing if the bought components fit or perform to the requirement. There were a lot of components which were rejected in this method, but the final result of such experiments gave us satisfactory results. The components which fulfilled our requirement are all included in the device. The discarded are to be regarded as research and development liabilities. Component wise costing of the device is given in Table 4.2.

Table 4.2 Costing Table.

COMPONENTS	QUANTITY	COST
1.Sunboard	3x4	240/-
2.Fevibond (Adhesive)	1 Tube	55/-
3.Sandpaper	1	10/-
4.Plastic Scale	1	12/-
5.Aluminium Composite	1x2	1600/-
6.Aluminium Sheet	2x3	350/-
7.Glue Stick	1	20/-
8.Ball Bearing	4	200/-
9.Welding	2 Times	270/-
10.Galvanised Metal	3x4	520/-
11.Fabrication	--	1170/-
12. WD-40	1	80/-
Total	43	4527/-

TOTAL COST = Rs. 4527

4.6 CONCLUSION

With the use of electronic devices, it becomes easier to use and also increases the efficiency of the incinerator. The operating procedure becomes much simpler. The device may now be kept on for a full day and it can be easily operated without any special skills.

CHAPTER 5

APPLICATION AND LIMITATION

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5.1 APPLICATION

This device can be used in places such as schools, colleges, public places, hospitals, railway stations etc. We designed and manufactured this device such that every organization big or small could afford it. It is designed such that a multi storey building to harness energy.

5.2 LIMITATIONS

Main Limitation of this device is that:

- Requires power and a starting motor to start the VAWT wind turbine
- Need guy wires to ensure the pole stays vertical so blade rotation is smooth
- Not as efficient as horizontal-axis wind turbines.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

If any conclusion can be drawn is that the VAWT undoubtedly will be with us in the future, and can be seen all around us, as has happened with other renewable technologies for electricity production, such as HAWT and PV, thus becoming part of the future renewable energy range and the business network, contributing to the reduction of CO₂ production and economic growth. Even after being a subject to which many studies have been devoted, however we still have a long road ahead and certainly there continue to be many areas to experience. That is why, after doing this article, the Department of Energy and Marine Propulsion of University of A Coruna, have determined a preliminary geometry for the development of a new model of VAWT, in which is working out, doing various tests using computational methods in order to obtain optimal morphology and even making a preliminary model prototype, thus doing their bit to the development of this technology.

6.2 FUTURE SCOPE

Lot of scope for improvement in this device such as:

- This designed turbine does not require more space.
- Because of low vibration structural damage is minimum so, we can place this turbine in ant governmental or residential or commercial building.
- If we are using high grade carbon fiber material we can install at high chimney, on a boat or at a border of country where electricity is not available.
- By setting different angles at different speed of the turbine can also be done as a future work or scope.
- Due to its low cost this will be very beneficial once manufactured on scale hence provide access to previously overlooked clientele like domestic users, farmers, sustainability enthusiast.

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Structural Optimization of composite wind mill blade with variation of density

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Abstract— In an effort to find solutions for global energy crisis, an analysis on a vertical axis wind turbine was conducted with the consideration of renewables and energy efficiency. Small scale vertical axis wind turbines have a number of advantages for deployment in an urban environment but are subject to highly varying thrust and radial aerodynamic forces. The project particularly aimed to address its use as the electricity supply for residential properties or any other places with less ideal economy conditions. The uncomplicated and highly accessible mechanism using basic materials is to give people to have a viable option on their own electricity production.

Keywords—vertical axis wind turbine, electricity supply, basic material, less ideal economy

I. Introduction

The consumption of fossil fuels has increased, resulting in high CO₂ emissions and serious climate change. Research on renewable energy is actively under way to solve these environmental problems and in anticipation of the depletion of fossil fuels[1] As the demand for the electricity is increasing day by day and its production was not able to fulfill the requirement. Thus, there is a need of alternative source of energy production, which is independent of the fact that the source of the energy should be renewable and non-polluting. A wind turbine produces no environmental harm during the production of energy. It consists of propeller, which rotates the electrical device for power generation.[2] The present study demonstrates the analysis of a vertical wind turbine used at highway dividers. When a vehicle passes near to the wind turbine, it produces the considerable amount of air due to speedy vehicle. This air strikes the blade of the vertical axis turbine and makes the turbine blades to rotate. The rotation of the turbine generates electrical energy. Furthermore, the air striking produces vibration, due to sudden impact of air on the blade. Many researchers contribute to design and analysis of wind turbine to improve the performance.[3]

The turbine rotates by the power of moving vehicle air impact and generates the non-polluting energy.

[4] Wind energy is a clean and inexhaustible energy source widely used as a working fluid for wind farms for centuries. However, its use as a means of electricity supply began modern era due to the rise of environmental concerns and fuel resources issues. The global demand for sustainable and renewable energy has created the necessity for

research and the development of new technology. Hence, the wind energy has been the focus of the industry and has considerably grown its use but just in a large-scale production. In recent years, significant increase of more efficient, larger, and expensive horizontal axis wind turbines (HAWT) appeared to create onshore and offshore wind-turbine fields.

[5] This study aims to produce electricity on a lower scale by using a small wind turbine in order to generate a house-hold electricity supply and build a cost-effective and accessible turbine for people who need an alternative option to cover their own electricity demand. This report presents the rotor blade design, and the results of the experimentation of a helical vertical axis wind turbine (VAWT). These turbines come with a few specific advantages over the horizontal ones, and those advantages make this kind of turbine a better option in a city or more challenging locations. Key features and its explication are described below:

- **Work speed range:** The VAWT starts to work under lower wind speeds (about 3 m/s) and can still work on high wind velocities (about 40 m/s).
- **Capturing all wind directions:** Such turbines can catch wind in all directions without requiring the heavy and expensive directional mechanisms.
- **Efficiency over higher turbulence levels:** The HAWT needs an axial wind incidence on the blades. Therefore, when wind turbulence increases, the efficiency decreases. However, VAWT catches the wind in all directions, so the change of wind directions, which is common in cities, does not affect its efficiency.
- **Turbine size:** The mechanism of VAWT is simpler and much smaller. HAWT can be over 30 m high and 20 m diameter. VAWT is approximately 3 m high and 1.5 m diameter.
- **Construction, maintenance, and transportation costs:** As mentioned the VAWT is simpler and smaller therefore, the construction and maintenance are not complicated and due to the smaller scale, the cost for transportation is significantly lower than

HAWT.

- Reduced noise pollution and visual disturbance level: This operating mode helps the helical vertical axis wind turbine make less noise (about 50 DB) and its twist geometry eliminates the flickering moving shadows casted by the blades.[6]

II. DESIGN CALCUALTIONS AND PARAMETER SELECTION

A. Wind Data Collection

To begin with the design calculations, data regarding the wind speed and density at the deployment location had to be collected. Various sources such as IMD, Wind Alert and Air Traffic Status were considered and the following data was obtained.

- Air Density (ρ): 1.275 kg/m³
- Average Wind Speed (v): 3.7m/s
- Air Viscosity: 1.48×10^{-5} N/m²s

The above equation shows the significance of wind speed (v) in power generation which increases comparably as wind speed rises to cubic power [7]

B. Power Calculations

Power that can be harnessed from the wind ' P_w ' is given by.

$$P_w = 1/2 * (\rho A_o v^3) \quad (1)$$

The above equation shows the significance of wind speed (v) in power generation which increases comparably as wind speed rises to cubic power [7].

Betz' Law states that any wind turbine can harness only 16/27th of the wind's kinetic energy.

Actual Mechanical Power that can be harnessed from the wind ' P_m ' is given by

$$P_m = 1/2 * 16/27 (\rho A_o v^3) \quad (2)$$

Using the above formulae, Swept Area (A_o) was found to be 0.0770m² or 770cm². Thus, the Rotor Diameter and Rotor Height were finalized to be 30cm and 70cm respectively.

C. Airfoil Selection

After consulting various journals and handbooks written on Wind Turbines, it was deemed necessary that an airfoil profile which generates lift is required for generating power. Typical design parameters for VAWT Blades are as follows:

- Often Symmetric or slightly asymmetric
- Light Camber for self-starting purposes (up to 4%)
- Thickness of 12% to 22% of the chord length one

The main objective of this research is to reduce the overall cost for which we selected NACA 2412 airfoil.

D. Reynolds number calculation

The Reynolds number is a dimensionless value that measures the ratio of inertial forces to viscous forces and describes the degree of laminar or turbulent flow. Systems that operate at the same Reynolds number will have the same flow

characteristics even if the fluid, speed and characteristic lengths vary.

Air Density changes with changes in temperature and with the varying pressure related changes for plain sailing we assume average air density at average temperature of 30°C 1.165 Kg/m³.

Average Wind speed in Navi Mumbai is 3.7m/s with maximum wind speed being 11m/s.

The Reynolds number is calculated from:

$$Re = (\rho v l) / \mu = (v l) / \nu \quad (3)$$

Where

v = velocity of air

l = the characteristic of length the chord width

ρ = density of the air

μ = dynamic viscosity of air

ν = the kinematic viscosity of air

With the help of above formula we found out that :

$$Re = 25924$$

E. Equations

The tip speed ratio (λ) is closely related to the power coefficient. The tip speed ratio is defined as the ratio of the blade tip speed and the wind speed at which the blade tip moves with rotation, as shown in Equation (4).

$$\lambda = R\omega / V \quad (4)$$

All wind turbine rotors have an optimum tip speed ratio with maximum power. The optimal ratio is related to the change of the incoming wind speed. The rotor swept area (A_w) is determined by the radius and height of the wind turbine.

$$A_w = 2RH \quad (5)$$

The windswept area should consider the height of the rotor (H) and the aspect ratio with respect to the radius (R). The longer the rotor radius, the higher the generated torque, but the longer the strut length, the lower the structural stability. However, when the rotor height is greater, the generated torque is lower, and the rotational speed of the rotor should be increased to obtain the same power output. The aspect ratio (AR) can be expressed as

$$AR = H/2R \quad (6)$$

Solidity (σ) is an important variable that determines the performance of wind turbines. Solidity is defined as the ratio of the total projected area (NC) of the rotor blade to the rotational area of the wind turbine rotor. The projected area is the projection in the vertical section of the rotating shaft and can be expressed as

$$\sigma = NC/2\pi R \quad (7)$$

The blade chord length (C) can be calculated using the

solidity. The chord length is the length of the airfoil and is an important design variable because the generated torque changes according to the chord length.

Urban wind power generators should operate at low speed with low noise. A Savonius wind turbine can rotate with a tip speed ratio of less than 1, but the vibration and noise are severe due to the characteristics of a drag-type rotor.[8]

Transmission Efficiency is assumed to be at 59.9% and the power coefficient is 0.593 maximum in HAWT this is also known as Beltz Limit, for VAWT the maximum CP is 0.4. We will assume that to 0.15 after considering various values from the research we concluded.

III. THEORETICAL CALCULATIONS

$$P = \frac{1}{2} \rho A w v^3 \eta CP \quad (8)$$

Where, ρ = Air Density

A_w = Area Swept

v = Wind Speed

η = Transmission Efficiency

CP = Power Coefficient

power output per unit swept area

$$P = (0.5 * 1.165 \text{ Kg/m}^3 * 1 \text{ m}^2 * (3 \text{ m/s})^3$$

$$* 59.9 * 0.15) 1000$$

$$= 0.1413 \text{ Kw/h}$$

Since there are 8760 hours per year, so the 0.1413 kw/h becomes 1237 Kw/h per year

Below table shows the power output at different velocities produced by 59.9% efficiency and 1m2 wind turbine in an ideal condition

Velocity (m/s)	Power Output (kw/h)	Power Output Per Year
3	0.1413	1237
5	0.6542	5730
7	1.7951	15725
10	5.2337	45847
11	6.6991	58684

IV. Material Selection of blade

The main motive of this experiment was to determine the most cost effective material to use with it trying to increase the efficiency of the blades by comparative method with trial and error, the main

parameter for selection was the density of the blade but with it other elements like weather proof , rigidity, less corrosive, availability, plasticity, brittle and malleable.

We tried with four materials in total while testing it on all the above criteria and those are as follows:

Material	Density
Galvanized Iron	7.809g/cm ³
Sunboard	0.33g/cm ³
Aluminum Composite	1.51g/cm ³
Aluminium	2.7g/cm ³

The most suitable material as it turns out was aluminum with a density of 2.7g/cm.

V.FABRICATION

In our prototype we created a blade from a rectangular sheet of height 70cm and width 22cm, we cut three such sheets.

These sheets needed shaping for which we hammered it with a wooden mallet on a smooth wooden surface so as to not create any ridges upon it, this was then folded from 10cm from one side.

Originally our plan was to put rib bits into the blade but this would have posed two major concerns

1. It would have damaged the structural integrity of the blade.

2. Rib bit would cause uneven distribution of weight thus, defeating the whole purpose of the blade.

To avoid this problem all together we came up with solution of creating a double seam :

The double seam is made using a double seamer, which can have just one or a number of heads or seaming stations. The double seam is formed by mechanically interlocking five layers of material together: three layers of the can end and two layers of the can body. Each seaming head typically consists of two rolls, a first operation roll and second operation roll, and a chuck. Some seaming machines have two first operation rolls and two second operation rolls and a few machines use a method called "rail seaming" which requires no rolls. During the seaming operation, the can end is lowered on to the filled can body and held down by the chuck, which acts as an anvil to the seaming operation. The first operation roll then engages the can end against the can body thereby folding the end curl around the flange of the body. In some seaming machines, this is done as the can is turning at high speed. In other seaming machines, the can is stationary and the first operation roll (or rolls) spins around several times to ensure a complete first operation. After the first

operation is complete, the first operation roll disengages from the can and the second operation roll then engages the can. The purpose of the second operation is to iron out the double seam into its final shape and remove the voids between the layers of can and end material. In practice, ironing out all of the can and end material in a double seam without leaving some voids is impossible without the use of a sealing compound.[9]

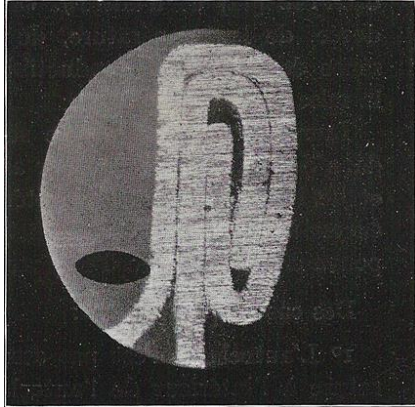


Fig.1

These blades then were attached to holder which was fixed in a rod, to increase the ability of the blades we gave it an angle of 30° hence increasing its attack angle. The whole blade assembly was the connected with a support structure to give it a safe height and stability but for its smooth functioning we needed to connect the structure and rotor assembly with a housing made by us which has two bearing at the end of a hollow cylinder pipe of which the diameter was chosen as such that it matches with the diameter of the rod.

VI. FUTURE SCOPE AND CONCLUSION

A helical vertical axis wind turbine was built with the purpose of covering the electricity supply for a household with an average consumption of approximately 1500 kW/h per year. The turbine was subjected under many tests to confirm its performance and operation ranges. After analyzing and comparing the results with the analytical data, it was determined that the helical wind turbine could be a viable alternative option for its use to generate cost-competitive energy. Wind power is a clean and inexhaustible source of renew able energy, which has experienced dramatic growth in the last decade. Considering the featured benefits, such as the construction and maintenance costs, turbine size and operation requirements, this rotor mechanism

could be a scalable solution, which has a significant expansion potential to address the current renewable energy demands.

We successfully created a working prototype which moves freely with wind speed as low as 3.5m/s.

This prototype was very cost effective which provides opportunity to people who wanted to own their own wind mill at a feasible price point.

Total amount required for the testing including all the fabrication was around Rs4500/- but to build this model with now attained knowledge would fall as low as Rs500/- if one were to scale this up the cost would be lowered further from buying raw materials in bulk.

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